Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 6 · SLO 1 — Solved Worksheet

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Unit â€“ 2 : Session â€“ 6 : SLO â€“ 1 | Singly Linked List â€” Insertion at the Middle

I. Insert a New Node VAL After a Node that has Value NUM

- Step 1: IF AVAIL = NULL
- Write OVERFLOW
- Go to Step 12
- [END OF IF]
- Step 2: SET NEW_NODE = AVAIL
- Step 3: SET AVAIL = AVAIL -> NEXT
- Step 4: SET NEW_NODE -> DATA = VAL
- Step 5: SET PTR = START
- Step 6: SET PREPTR = PTR
- Step 7: Repeat Steps 8 and 9 while PTR -> DATA != NUM
- Step 8: SET PREPTR = PTR
- Step 9: SET PTR = PTR -> NEXT
- [END OF LOOP]
- Step 10: SET NEW_NODE -> NEXT = PTR -> NEXT
- Step 11: SET PTR -> NEXT = NEW_NODE
- Step 12: EXIT

II. Insert a New Node VAL Before a Node that has Value NUM

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- [END OF LOOP]
- Step 10: SET PREPTR -> NEXT = NEW_NODE
- Step 11: SET NEW_NODE -> NEXT = PTR
- Step 12: EXIT